Alexander Bowring

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Summary

Senior Mathematical Consultant and Technical Lead with extensive energy sector experience, leading award-winning ML and optimisation projects for the National Energy System Operator and Scottish and Southern Electricity Networks across a **£2M+ project portfolio**. Lead end-to-end design and implementation of large-scale forecasting and optimisation models within cross-functional teams, utilising AWS and MLOps practices for model training and validation and mentoring junior developers to establish best practices. Regularly communicate operational value from complex analyses to diverse stakeholders, collaborating with project managers and senior stakeholders to ensure successful delivery. My extensive practical ML and optimisation experience, combined with a rigorous foundation in statistics and mathematics from my **PhD in Population Health (Oxford)** and **First-Class BSc Mathematics (Warwick)**, enables me to confidently build innovative solutions where established approaches don't exist.

Professional Experience

Senior Mathematical Consultant (previously Mathematical Consultant) — Oxford, UK

Smith Institute — *Nov 2021 - Present*

- **Technical lead** across several cross-functional teams of 5-8 members, delivering **£2M+ cumulative project value** through advanced ML and statistical solutions for the National Energy System Operator (NESO), Scottish and Southern Electricity Networks (SSEN), and Coca-Cola, with regular stakeholder communication across technical, commercial, and client audiences.
- **Technical lead** for the Dynamic Reserve Setting (DRS) project with NESO, a multi-year initiative designing and developing explainable gradient-boosted ML forecasting models for predicting national energy reserve requirements. Led the technical team in all aspects of data ingestion, model training and validation, managing and registering hundreds of config-driven experiments using AWS, and developed a Power BI dashboard enabling control room engineers to interact with model outputs. The DRS models, now deployed in the NESO control room, substantially outperform previous operational models, reducing energy reserves by gigawatts daily and **saving millions of pounds monthly** in reserve procurement costs. Recognised with the **Operational Research Society Presidents Medal Award 2024** for the best practical application of operational research.
- **Technical lead** for the Dispatch Decision Intelligence project with NESO, enhancing mixed-integer optimisation performance through implementation of warm start techniques and heuristics for the optimisers used within the NESO control room for generator dispatch decisions. Led the team in developing a tailored explainable optimisation module and interface enabling control room engineers to understand why the optimisers made specific generation dispatch decisions. Delivered significant improvements in optimiser run-times and decision-making performance, aspects of which have now been deployed in the control room.
- **Principal model developer** for the Vulnerability Future Energy Scenarios (VFES) project with SSEN, designing and implementing explainable gradient-boosted models for consumer vulnerability analysis. Used AWS and MLOps practices to perform modelling experiments, optimising model features and hyperparameters for predictive performance, then applied clustering methods on machine learning importance metrics to identify common vulnerability drivers across communities SSEN serves. Delivered actionable insights that enabled SSEN to optimise their consumer support strategy for vulnerable customers. Recognised with the **Utility Week Unlocking Data Award 2024** and the **DataIQ Award for Best Use of AI for the Public Good 2024**.

Early Career Research Fellow in Neuroimaging Statistics — Oxford, UK

Nuffield Department of Population Health, University of Oxford — *Nov 2019 - Nov 2021*

- Led research on variability in functional neuroimaging results caused by differences in analytic workflows, using Python, Matlab, and Unix to design and implement **parallel analysis on fMRI time-series data** to run **59 unique neuroimaging pipelines** on a high-performance computing cluster. Applied quantitative metrics including correlations, Euler characteristic, and DICE coefficients to comprehensively assess variability in statistical results due to different preprocessing and hypothesis-testing methods.
- Organised and led a weekly **Neuroimaging Statistics Oxford reading group** featuring international experts.
- Taught PhD students enrolled in the university's doctoral training centre programme statistical methods, programming skills, and aspects of independent research.

Research Assistant in Neuroimaging Statistics — Coventry, UK

Warwick Manufacturing Group, University of Warwick — *Nov 2015 - Oct 2016*

- Developed standard practices for data-sharing and meta-analysis in neuroimaging, improving reproducibility and collaboration.
- Performed extensive fMRI analyses using major neuroimaging software and maintained version control with GitHub, sharing results via online repositories.
- Mentored a visiting PhD student on a short-term research project that was **subsequently published**.

Undergraduate Research Intern — Coventry, UK

Undergraduate Research Support Scheme, University of Warwick — *Jun 2015 - Oct 2015*

- Assisted in developing software to visualise fMRI results and tested research prototypes through data analysis.
- Gained hands-on experience in collaborative research and effective communication with supervisors and team members.

Education

Doctor of Philosophy in Population Health — Oxford, UK

Nuffield Department of Population Health, University of Oxford — *Oct 2016 - Nov 2019*

Thesis: On the Reproducibility and Interpretability of Group-Level Task-fMRI Results

- Developed a novel **Confidence Sets method** for group-level inference on task-fMRI effect-size maps, contributing original statistical theory for linear modelling and bootstrapping methods. **Published two papers in NeuroImage** demonstrating ability to translate theoretical advances into practical applications, validating the method on large-sample fMRI studies.
- Evaluated reproducibility of task-fMRI results across multiple neuroimaging software packages, designing automated analysis pipelines and implementing statistical models for hypothesis testing on task-fMRI time-series data. Work ranked in **top 5% of research outputs by Altmetric**, demonstrating significant methodological impact.
- Processed and analysed big neuroimaging datasets, including the **UK Biobank and Human Connectome Project dataset**.

Bachelor of Science in Mathematics — University of Warwick, UK

University of Warwick — *Sep 2012 - Jul 2015*

Graduated with **First Class Honours**, achieving >80% in modules including Algebra II, Analysis III, Galois Theory, Programming for Scientists, and Mathematics by Computer.

Publications

I have several publications in high-impact journals including *Nature*, *NeuroImage*, and *Human Brain Mapping*. Full list available at Google Scholar.